

Document Variables (`_vars`)

When creating a report yaml document, you have the option to include document-level variables by using the '`_vars`' key in the root of the document.

These variables can be rendered as part of your strings (using jinja) within the report content using the double-braces (e.g. `{{variable_name}}`). When rendered using jinja, all values are stringified and only the string representation is shown (i.e. the result of `str(var)`).

This document has five variables defined, `a`, `b`, `c`, `d`, `e`.

- `a = hello`
- `b = True`
- `c = 23`
- `d = -0.013`
- `e = {'key': 'value'}`

Passing Python Objects

It is also possible to pass variable data as Python objects to blocks (see "YMPrint blocks") using the `$var` syntax (similar to bash).

You cannot use the dollar-sign variable name syntax (e.g. `$varname`) within content that is to be displayed (if attempted, an error will be raised). The intended use-case is for passing data within the document to be used as *block parameters*.

For example, you might want to have some consistent spacer used throughout your document. Instead of typing `"_spacer: 5"` at all locations, you can instead set a variable in the `_vars` to something like `"small_space: 5"` and employ the variable at each spacer as such: `"_spacer: $small_space"`.

The data you set in the `_vars` key can be any of the native YAML datatypes: `str`, `int`, `float`, `null/None`, `bool`, `list`, and `dict`.

Computing new Python objects within code

It is also possible to use a `_py` code block (see "YMPrint blocks") to create variables of arbitrary classes/datatypes and pass that data to block parameters.

A ready example of this would be the `_matplotlib` block which has a `"fig"` parameter that expects a matplotlib figure.

You would first generate the figure in a `_py` block and assign the generated figure to a variable. You would then use the dollar-sign syntax to pass that computed variable into the `"fig"` parameter of the `_matplotlib` code block (see YMPrint blocks for details.)

